

# 1ST CENTURY

## THE DATE BY WHICH CUNEIFORM SCRIPT HAD BECOME EXTINCT

Cuneiform (wedge-shaped) script developed from the earliest writing, invented in Mesopotamia around 3300 BCE. It was used for a wide range of ancient Near Eastern languages, including Sumerian and Akkadian.

Boat-building also developed significantly around the 3rd millennium BCE. Early humans had probably been using some form of boat from as long ago as 50,000 BCE, although the earliest surviving water craft is a dugout canoe that dates from around 7200 BCE. In the Gulf region, boats were being made of bitumen-coated reeds as early as 5000 BCE.

By around 3000 BCE, more sophisticated vessels made of wooden planks that were sewn together were being built in Egypt. Early boats were powered solely by oars. **Sailing boats, with square-rigged sails,** appeared in Egypt in around 3100 BCE, supplementing muscle power with wind power. By 3000 BCE, large steering oars

had been developed in Egypt, and by about 2500 BCE, pairs of **side oars and tillers** had been introduced.

Before 3000 BCE, few monumental structures were residential. The practice of building some temples on platforms began before 4000 BCE, the platform rising with each rebuilding. After 2900 BCE, temple platforms in Sumerian cities such as Ur and Kish reached a considerable height, leading to the **development of ziggurats**—initially three-tiered structures with a shrine on the top platform. Largely made of mud bricks, with a baked-brick facing, these monumental structures suggest growing sophistication in structural engineering.

In Egypt, **most architecture was religious (temples) or funerary (tombs).** The tombs of the nobility and rulers of the Early Dynastic Period (around 2900 BCE) were simple mud-brick rectangular structures known as **mastabas**. Between 2630 and 2611 BCE, during the reign of the pharaoh Djoser, a huge mastaba was

### Egyptian faience

*This Middle Kingdom (1975–1640 BCE) statue of a woman with a tattooed body shows the deep blue color typical of much Egyptian faience work.*



modified by building six stepped platforms to create a **step pyramid**. By the reigns of Khufu, Khafre, and Menkaure in mid-3rd millennium BCE, the creation of **smooth-sided stone pyramids** had been perfected, and each of these pharaohs erected a huge pyramid tomb for himself at Giza. Collectively known as the **Great Pyramids**, each was oriented and built with great precision, which suggests that sophisticated surveying techniques were in use.

An interest in observational astronomy arose early in Mesopotamia, culminating in the **Venus tablet of Ammi-Saduqa** (dating from around 1650 BCE). It contained the rising and setting times of the planet Venus over a period of 21 years. A carved piece of mammoth tusk found in Germany, dating from about 32,500 BCE, may possibly represent the constellation Orion, but **systematic division**

**of the sky into constellations** dates from Babylonian manuscripts c.1595 BCE.

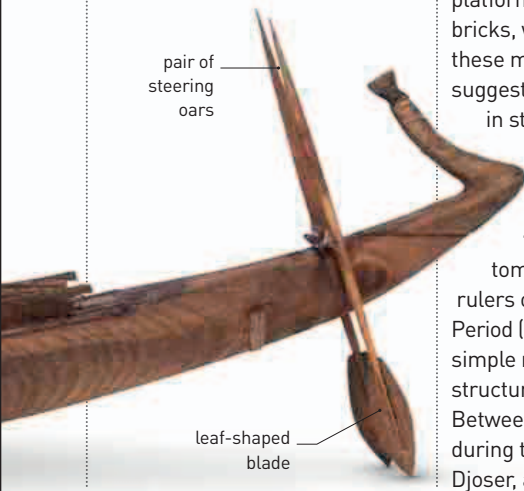
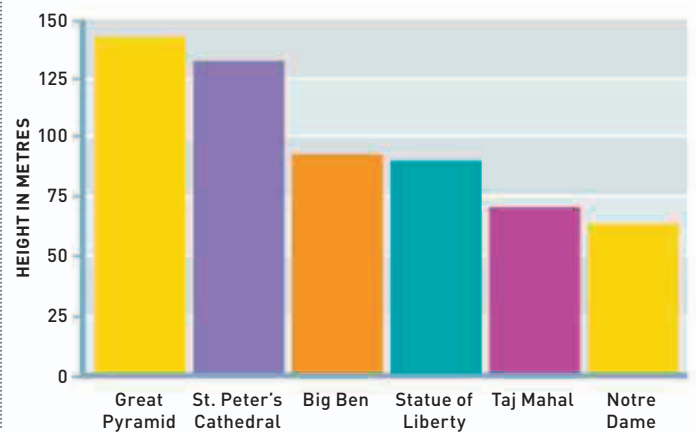
With the growing administrative demands of cities in the 3rd millennium BCE, the development of an **accurate calendar** became vital. The first known version is in the Umma calendar of Shulgi, a Sumerian document dating from about 2100 BCE that contains **12 lunar months of either 29 or 30 days**. When this 354-day year became too out of phase with the real 365.25-day year, an extra month was added by royal decree. The ancient Egyptians had a similar calendar, but five days were added each year to give a 365-day year.

There have been claims that some prehistoric carvings represent **topographical maps**, but **true cartography** and real maps were not developed until

the 3rd millennium BCE. The Akkadian **Ga-Sur tablet** (dating from about 2500 BCE) shows the size and location of a plot of land between two hills and was probably part of a land transaction. Fragments of a statue of Gudea of Lagash, from around 2125 BCE, show a plan of a temple. The **first real street map** discovered to date shows a scale plan of the Sumerian town of Nippur (in present-day Iraq) and dates from about 1500 BCE. The first surviving attempt to map the entire known world is the Babylonian “world map” from about 600 BCE, which shows the regions surrounding Babylon (see 700–400 BCE).

### Monumentally tall

*Built around 2560 BCE, the Great Pyramid of Khufu was 482 ft (147 m) tall and remained the world's tallest building for nearly 4,000 years.*



**c.2500 BCE** Babylonians produce the Ga-Sur tablet, the **first true map**

**c.2500 BCE** Mesopotamians invent **gold granulation** process

**c.2500 BCE** Egyptians use pairs of **side oars and tillers** on boats

**c.2500 BCE** Erection of **Stonehenge** stones begins in England, a megalithic monument that indicated important solar and lunar events

**c.2400 BCE** Mesopotamians invent the **shadoof**, a device for raising water

**c.2200 BCE** First **ziggurats** are built in Mesopotamia

**c.2100 BCE** Sumerians produce the **earliest known calendar**, the Umma calendar of Shulgi